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HUMAN CHAT AND SELF-ORGANIZED CRITICALITY: A CHANCE DISCOVERY APPLICATION

CALKIN A. S. MONTERO*
Graduate School of Information Science and Technology
Hokkaido University, Kita-ku Kita
14-jo Nishi 9-chome, 060-0814 Sapporo, Japan
calkin@media.eng.hokudai.ac.jp

KENJI ARAKI
Graduate School of Information Science and Technology
Language Media Laboratory, Hokkaido University
Kita-ku Kita 14-jo Nishi 9-chome
060-0814 Sapporo, Japan
araki@media.eng.hokudai.ac.jp

It has been said that there is something deeply touching about creating something and then having a *chat* with it. Machines in general, and more specifically computers, are not the exception to that tendency. In fact, since the Turing dissertation "Computing Machinery and Intelligence", machines' ability to perform a human-like conversational behavior has been regarded as the basis to consider the machines as "intelligent entities". In this paper, we propose the idea of the discovery of critically self-organized interaction, using its inherent structure as the basis to modeling computer chat.

Keywords: Critically self-organized chat; chance discovery; natural language processing.

1. Introduction

Machine conversation with humans, as an area of research and development, has recently made exponential advances nowadays. Since Alan Turing's philosophical dissertation in 1950[1] regarding the assumption of judging machines as "intelligent entities" by means of dialogue, the development of conversational machines has increased highly. In the 1960s, many of the issues raised by Turing became relevant in the context of one of the most famous examples of human-computer conversation (HCC) systems: ELIZA.[2] ELIZA is pattern-matching natural language program whose goal is to assume the role of a Rogerian therapist during its dialogue with the user. The dialogue management of the program is modeled through a cascade of regular expressions substitutions, each one matching some part of the user's input lines and changing them into a system answer.

*Present address of the author: Kita-ku, 001-0015 Sapporo, Japan.

Another well-known dialogue program is PARRY.[3] While ELIZA simulated a psychotherapist, PARRY attempted to simulate a paranoid schizophrenic patient. The program implemented a model of the behavior of a paranoid patient based on concepts, conceptualization and beliefs — i.e. judgments about conceptualizations: accept, reject, neutral. The program also possessed an explicit conversational strategy:

(a) Notion of the topic: trying to determine *Flare concepts*,[a] i.e. any concept close to its *delusional network*. This network were based on concepts involving the Mafia controlling crooked gambling at the track.
(b) Determination of other's intention: *malevolent, benevolent or neutral*. In order to accomplish this goal, *grammar* of malevolence and benevolence detection rules were implemented. For example, the conceptualization "are you sure?" is classified as an *implicit insult*, hence a form of malevolence. In the same regard, "you are right" is a kind of *agreement*, hence a form of benevolence.
(c) Emotional reaction to the detection of malevolence: psychological harm increase anger, physical treats increase fear, references to delusional network increases fear, as a result increasing mistrust. Benevolence reduces negative emotions. In summary, simulation of emotional reaction to what was said and alteration of its behavior accordingly.

PARRY introduced some important ideas for the development of *conversational agents*: context and topic *understanding* and conversational strategy or dialogue management.

Modern examples of conversational agents include airline travel information systems,[4] speech-based restaurant guides,[5] and telephone interfaces to email or calendars.[6] The dialogue manager is the component of such conversational agents that controls the flow of the dialogue, deciding at a high level how the agent's side of the conversation should proceed, what questions to ask or statements to make, and when to ask or make them.

Three main approaches have been proposed to modeling dialogue[7]: dialogue grammars approach, plan-based approach, and joint action or collaborative approach. The dialogue grammar approach is based on the observation that sequencing regularities appear in the dialogue[8] — so that questions are generally followed by answers, proposal by acceptances, etc. Hence, it has been proposed that dialogues are a collection of such act sequences with embedded sequences of digressions and repairs. This kind of grammar can be constructed following the Chomsky hierarchy or finite state machines. The plan-based approach is based on the observation that the user's utterances are not simple string of words but communicative actions or speech acts.[9] This approach assumes that the user's speech acts are part of a plan and the goal is to uncover and respond appropriately to it.[10] Finally, the joint action or collaborative approach views the dialogue as a collaboration between the

[a] *Italics* are used for emphasis.

interlocutors in order to achieve a mutual understanding.[11] All of those approaches focused on creating a dialogue in order to achieve certain specific goal, that is to say, domain specific dialogue approaches.

Recently, the development of non-specific-goal[b] dialogue agents has exponentially increased. Those agents are called chat robots or chatbots. A chatbot is defined as a computer program that simulates human conversation through Artificial Intelligence (AI). In fact, ELIZA has been considered the first chatbot. The dialogue modeling of a chatbot is based on certain word combinations that it finds in a phrase given by the user, and in the case of sophisticated chatbots like ALICE,[c,12] AIML[d] is used to transcribe pattern-response relations. Chatbots are widespread on the Web, being applied for electronic commerce customer service. A chatbot application to question answering systems has been proposed by C. S. Montero and K. Araki.[13–15] However, it is highly improbable to create all the possible patterns-response that could appear during a dialogue, a shortcoming that makes the chatbot to easily lose the conversation flow.

By analyzing the structure of chats, we realized that the dialogue could be interpreted as a critically self-organized system, obeying the Zipf's law, where a utterance could contribute to the continuation of the dialogue flow or could crash the interaction. By discovering what are the more likely flows a conversation could have, and then creating the appropriated utterances based on the obtained knowledge, we believe a smooth, more human-like machine chat could be achieved. Critical utterances in a conversation — utterances that could change or completely crash the flow of the dialogue — could emerge by applying Chance Discovery techniques.[16] In this paper, we present the fundamental idea of modeling chat by discovering the more likely flows in a conversation, becoming aware of the criticality of the chat and applying a Chance Discovery tool like *KeyGraph* for analyzing the chat — critical utterances.

2. Critically Self-Organized Chat (CSOC)

In this paper, we consider a chat session to be a critically self-organized system, that is to say Critically Self-Organized Chat (hereafter abbreviated as CSOC). However, before immersing into the definition of those terms, let us state the general basis of critically self-organized systems.

Critically self-organized (CSO) systems[17,19] as a scientific field of study is relatively new although the basic questions about how organization arises have been brought up since ancient times. In this regard it has been observed that many natural systems show organization, e.g. galaxies, planets, chemical compounds, cells, organisms and societies. It has been stated that the complex nature's behavior shows

[b]Here non-specific-goal means open domain dialogue.
[c]Artificial LInguistic Computer Entity.
[d]Artificial Intelligence Markup Language, base on eXtended Markup Language, XML.

the tendency of large systems to evolve into a poised, "critical" state, way out of balance, where minor disturbances may lead to events of all magnitude. Most of the changes occur through catastrophic events rather than by following a smooth gradual path. Therefore, it can be said that complex phenomena observed everywhere indicate that nature operates at the self-organized critical state. Complex phenomena emerge when elements are interacting and connected into systems, as it is the case of the grains of sand in a sandpile.[17] The critical state has been said to be the point where the system properties could change suddenly. In the case of the sandpile paradigm, by adding grains of sand to the pile, eventually it reaches the state in which the addition of one single grain of sand can crash the whole pile, and at this point, it is said the pile has reached *the critical state*. In essence, a complex system is a functional whole, consisting of interdependent and variable parts. Complexity Theory states that critically interacting components self-organize to form potentially evolving structures exhibiting a hierarchy of emergent[e] system properties.

Generally, the field of self-organization seeks general rules about the growth and evolution of systemic structure, the forms it might take, and finally methods that predict the future organization that will result from changes made to the underlying components. The results are expected to be applicable to all other systems exhibiting similar network characteristics. It has been stated[16] that "in systems where the global state depends in a crucial way on the action of the individual components, it is clear what can be considered a 'chance'. A possible action which might appear at a certain moment in time as a chance for a single member of the system may turn out to alter the entire configuration of that system, leaving the individual members in a very different situation than anticipated."

Being introduced the reader to CSO systems basic notions, let us explain the concept of CSOC by using an analogy with the canonical example of CSO systems: the pile of sand, as described by Bak in Ref. 17. In the sandpile paradigm, grains of sand are being added so as to create a pile. "In the beginning, the pile is flat but as time goes on the sand slides become bigger and bigger. As the pile becomes steeper small sand slides or avalanches occur. The addition of a single grain of sand can cause a local disturbance, but nothing dramatic happens to the pile... Eventually the slope reaches a certain value and cannot increase any further, because the amount of sand added is balanced on average by the amount of sand leaving the pile by falling off the edges... There will occasionally be avalanches that *span*[f] the whole pile. This is the self-organized critical (SOC) state."

Let us consider our system a trivial conversation — chat — between two interlocutors. In the beginning the chat could be considered flat (greetings from each interlocutors). As the chat evolves toward one direction — some specific topic — certain utterances may cause the course of the chat to slightly chance the direction

[e]The appearance of a property or feature not previously observed as a functional characteristic of the system.
[f]Emphasis provided.

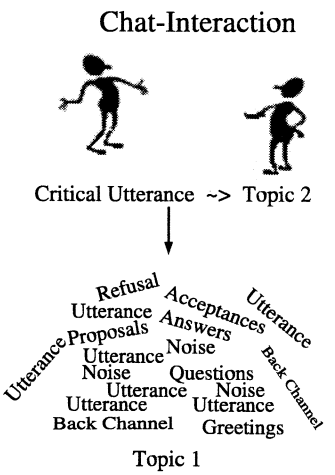


Fig. 1. A chance that exists at the cross point of scenarios. The scenario between the arrows emerges from Scenario/Topic 1 and Scenario/Topic 2.

of the topic, although the initial topic still remains the same. Eventually, during the chat there is nothing else to utter about the initial topic, so this subject cannot be discussed any further, either because the interlocutors have already agreed in their ideas regarding the subject or because the topic is no longer of interest for the interlocutors. At this point a single utterance given by any of the interlocutors can cause the whole direction of the chat to change toward a new topic — span of the sandpile — and the whole process starts again. Therefore, we call this phenomenon *Critically Self-Organized Chat (CSOC)* (see Fig. 1). Let us explain this phenomenon with an example of a fragment of a real chat between two friends.[g,18]

1A: Hello
1B: Hello A, ah
2A: Yes, B ah
2B: Ah
3A: Have you had your lunch
3B: No
4A: Uhm I haven't had my lunch also
4B: Uh busy uh
5A: Uhm busy playing with my nieces
5B: Both of them are in their own place eh at your home now
6A: Hah The younger one is always at my home mah
6B: Both of them – Orh
7A: My mum look after her
7B: Ah
8A: Then the older one ya lah today my sis you know brought her here lor
8B: Orh so you're having a fun time
9A: Uhm okay lah I mean uhm it's not every day that I can play with little children lor
9B: Uh huh but very tiring right – Playing with them
10A: Hah
10B: Can be very tiring
11A: With the older one lor – Get you to do a lot of things for her
11B: Ya
12A: She like to play slide you know

[g]The speakers names have been omitted.

12B: Uh huh
13A: So you must you know use your leg then let her slide down – So that is tiring lor
13B: Really aching legs
14A: Okay lor I haven't been exercising mah – So that's my exercise
14B: How big is the [unclear] word [/unclear]
15A: Uh March will be two years old lah
15B: Oh two years so now one plus almost two uh
16A: Almost two lor
16B: Orh quite heavy you know
17A: Then the the younger sister only two three months old ah
17B: Uhm uhm uhm
18A: The baby is only two three months
18B: Um so weekends the parents come over lah
19A: Ah so weekends supposed to fetch the the little baby home lor
19B: Orh
20A: My mum look after for weekdays ah
20B: Uhm they'll come and fetch her lah
21A: Ah they'll come and fetch – Ya they'll come and fetch her correct – So for this weekend it's different because the parents intend to leave the the big big girl here
21B: Orh
22A: [unclear] word [/unclear] good lor
22B: Uhm uhm uh uh
23A: I mean we can play with her lor
23B: So you've got entertainment
24A: [unclear] word [/unclear] Then she like to go out and all that lah
24B: Ah huh
25A: She's quite quite playful lah
25B: Ah huh
26A: Uhm
26B: Uhm you can
27A: Quite active – Huh
27B: Ah huh you can nuh handle her uh
28A: Can – Hey she's very attached to me leh
28B: Ah
29A: More attached to me than to my mother lah
29B: Orh
30A: Ah
30B: You spend a lot of time with her lah before
31A: Ya because let her slide down my leg and all that sure she like me lah
31B: Uhm not bad nuh – Gain experience first
32A: Cheh ya because ya at first I don't really like dare to carry the young one nor that means the two two month old baby ah
32B: Ah
33A: After a while orh can nah I dare to carry her dare to swing her about also
33B: Uhm my brother's wha the two sons oh very naughty
34A: Oh I know Nicholas and who ah
34B: Heh the other one doesn't have got a Christian name
35A: Oh doesn't have ah
35B: So one is boy boy – One is Wei Wei – Ah very tedious
36A: So when they grow up they'll wonder right – How come one
36B: Ya
37A: one have one don't have
37B: Ya ya ya don't know why Uhm
38A: Different nuh
38B: But ya lor
39A: You hear my little niece crying
39B: Heh uh Uhm so are you spring cleaning already
40A: Ah intend to do it today lor
40B: Orh
41A: So I will be home whole day lah but
41B: Uhm
42A: Uhm you know like can't get started like that do a little bit then no don't want to do uhm then
42B: Uh huh uh huh
43A: Basically it's ya lah – I mean because like weekends suppose to do all my laundry one mah
43B: Ah hah
44A: So I must kau tim my laundry first
44B: Uh huh
45A: Then ya lor then clear clear the table and all that lah

45B: Orh orh – Clear the table Oh your own table ah
46A: Ya lah my own table
46B: Orh orh – Ya ya ya
47A: You know a lot of things would have stacked up
47B: Yes – Same here
48A: So I very slow one mah – Ya just do a little bit then you know go and do something else you see
48B: Ah Aiyah – Can't get myself to start it also uhm uhm
49A: But aiyah have to do it today lah because it's like so near already
49B: Uhm
50A: left one and a half weeks only
50B: Next weekend
51A: [unclear] word [/unclear] weekend just one just over a week more only what
51B: Ya just over a week
52A: Ah – Then weekdays definitely cannot – Weekdays I'm so busy in the office you know
52B: Ah hah
53A: This whole this week itself ah
53B: Ah hah
54A: Every night you know
54B: Ah
55A: Past nine o'clock then I leave the office
55B: Whoa everyday
....
....

In this chat, the utterances 1A to 4B form the flat initial state of the system: greetings and topic-introductory utterances. At this point if there are no big changes in the chat, it could be said that the system is in equilibrium. The next utterance — 5A — initiates the chat toward a determined topic: A's nieces. This topic remains the main one until the utterance 33A. At this point a small "avalanche" occurs: the utterance 33B, changing slightly the topic from A's nieces to B's cousins until utterance 39A, yet the main topic remains. However, it is worth noticing how utterance 39B "crashes" the main topic (catastrophic avalanche) initiating a completely new topic: "laundry on weekends" — this is called *"a critical utterance"*. There are also small avalanches within this new topic — utterances 45A and 48A, yet the main topic remains. Then utterance 52A "crashes" the previous topic and started a new one: "busy at the office."

It is worth mentioning how every time an utterance crashes a topic, the system (chat) grows from an initial flat state (equilibrium) to a state where the addition of a single utterance can change the direction of the topic (catastrophic avalanche), initiating a new cycle.

We believe that understanding this human conversational behavior is useful when modeling computer-chat. That is because if the computer is able to smoothly follow the flow of the chat it could achieve a human-like conversational behavior. However, the question remains: how to provide the computer with such a knowledge? We intent to give an answer to that question through Chance Discovery, as detailed in Sec. 3.

3. Discovering Critical Utterances

In Sec. 2, it was explained how a chat rises from an initial flat state to a state where a single utterance can completely change the direction of the conversation, introducing a new topic, which follows the same loop as the previous topic repeating the cycle

once and again. Considering the flow of a chat, it can be said that the chat is a dynamic system whose environment is formed by the interlocutor's utterances. In this dynamic system, a *chance* could be seen as the moment when a critical utterance may come so as to define the new direction of the chat. Therefore, identifying critical utterances is crucial for the smooth flow of the chat. Having the knowledge of what a given topic's critical utterance may be like, modeling computer chat could be greatly improved. Here is where data mining or statistical techniques could be applied. In this regard, particularly *Chance Discovery* techniques are coming handy because of their emphasis on rare and possibly opportune events, considering the awareness of a critical utterance where an opportune giving birth to a new topic to focus during the chat.

A chance has been defined by Ohsawa and McBurney[20] as "a piece of information about an event or a situation which significantly impacts on the decision-making of humans, agents, and robots". From this perspective, discovering critical utterances may be crucial in developing computer chat models when the goal is a human-like chat flow.

3.1. *Critical chat and power law*

It has been proved that critically self-organized systems dynamics tend to obey a power law function[17] whose plot tends to become a straight line. The power law, also known as Zipf's law[21] after the Harvard linguistic professor George Kindsley Zipf, is the observation that frequency of occurrence of some event F, as a function of the rank r when the rank is determined by the above frequency of occurrence, is a power-law function $F_r \approx r^{-a}$, with the exponent a close to unity. Professor Zipf observed some simple regularities in systems of human origin, e.g. the population of a city as plotted as a function of the rank (the most popular city is ranked number one, etc.) is a power-law function with exponent close to 1, being the curve a roughly straight line on a logarithmic plot; the frequency of English words in a piece of literature when the number of occurrences is plotted as the function of the rank (1, 2, 3, etc.), the functional form is a power-law function with exponent close to 1. Zipf made similar plots for many geographical areas and found the same behavior.

In order to observe the self-organized critical behavior of the dynamics of the chat, the frequency of the used words during the chat showed in Sec. 2 has been analyzed.

Applying the Zipf's law to term occurrences in a collection, it has been expressed as roughly fitting the relation $r_t * F_t = K$. The probability that a randomly chosen term (with frequency F_t) will have rank r_t is given by $P_r = F_r/N$ where N, is the number of words in the collection. Table 1 shows the fifteen more frequently used words during the chat — most frequent word rank 1, and so on. Due to the nature of the data some of the words were combined so as to be considered the same — e.g. "yes", "yeah" and "ya", were counted as "yes"; and other expressions were not considered, e.g. "lor", "Orh".

Table 1. Frequencies of words during the chat: 476 total word occurrences, 215 unique words.

Frequent Word	Frequency	Probability
ah	64	0.134
you	63	0.132
uhm	61	0.128
I	46	0.097
lah	41	0.086
yes	40	0.084
the	38	0.080
so	36	0.076
know	31	0.065
uh	26	0.055
and	24	0.050
that	21	0.044
okay	19	0.040
but	15	0.031
we	14	0.029

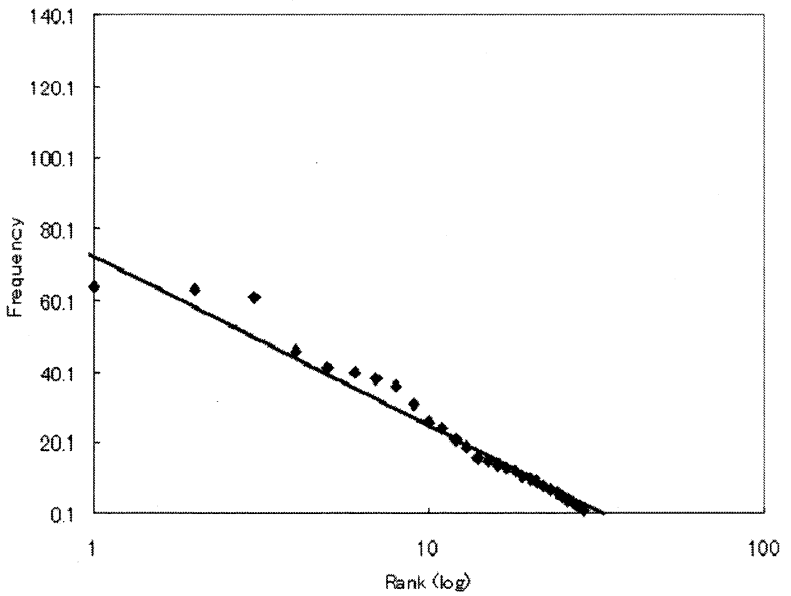


Fig. 2. Frequency as a function of the rank for the given chat. The curve tends to become a straight line with slope −3.64. The chat can be considered as a critically self-organized system.

Plotting the number of occurrences of a word as a function of the rank for the showed chat, the obtained graph is presented in Fig. 2. It is worth noticing that despite the small database (only one chat section was analyzed, 13 KB, 311 turns during the chat) the plotted graph shows a tendency to become a straight line with

a slope of −3.64 and the words that appears only once during the chat represent 45.2% of the total word occurrences. It can be considered that the slope of the curve is not approximately 1.1, as it is the case of the English text, and most critically self-organized systems analyzed using a large database, because of the limitation of the analysis data. Analyzing a larger database, the tendency of self-organized criticality can be better observed. However, despite the small database analyzed, the tendency of the chat to critically self-organize can be appreciated.

3.2. *Designing experiment: The KeyGraph*

In order to understand the behavior of possible chance events, a visual computer aid is needed. In this regard, the *KeyGraph* has been created as a data-mining tool for extracting patterns of the appearance of chance events. The *KeyGraph* tool was originally used for indexing documents.[22,23] As a data-mining algorithm, the *KeyGraph* identifies relationships between terms in a document particularly focusing on co-occurrence relationships of both high-probability and low-probability events. The output of the data analyzed by the *KeyGraph* is formed by a graphical view of clusters of co-occurring frequent events, being frequent events represented by black dots, and interconnecting infrequent events, represented by red dots.

Since human–human chat reflects the tendency to criticality self organize, as shown in Sec. 3.1, we defined *a chance* as the moment when shifting topics occurs within a conversation (cross point of scenarios). Therefore, it becomes essential to understand how criticality rises within topics prompting their shifting during the chat.

As a fundamental experiment, in order to graphically visualize how criticality rises — relationship of the expression during each topic of the chat within crashes — we applied the *KeyGraph* to the chat example given in Sec. 2. The analyzed chat fragment contained three distinct crashes whose visual results can be appreciated in Figs. 3 and 4. As a matter of achieving a better understanding of the complex interrelations that exist between utterances during the chat, the experiment was designed for analyzing the first crash alone, so as to observe how a crash develops within the chat, and then the second and third crashes together with each other, as to observe the relationship between crashes.

In Fig. 3, the chat is analyzed from the utterance 1A to the utterance 39B. The main cluster in this figure, formed by black dots, is holding the central topic during this first crash: "little-baby-play". As shown in this figure, the main topic is surrounded by less frequent but interconnected words (grey dots), as it is the case of the interrelations "weekend-parents-come", "mum-look-after" and "slide-down-leg". In this graphic view the relationship between the main topic and its surroundings can be clearly seen. In Fig. 4, the chat is analyzed from the utterance 39B to the utterance 55B. In this figure two crashes are visualized: "laundry-basically-weekends" and "busy-office-definitely-cannot". Moreover, a small avalanche is well visually represented by an independent cluster. The structure of the crashes is

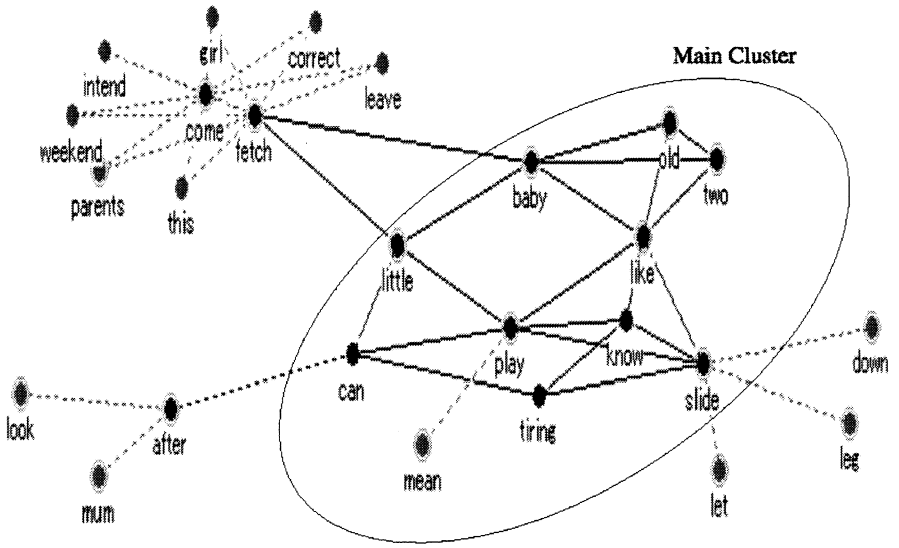


Fig. 3. First crash: Graphic view.

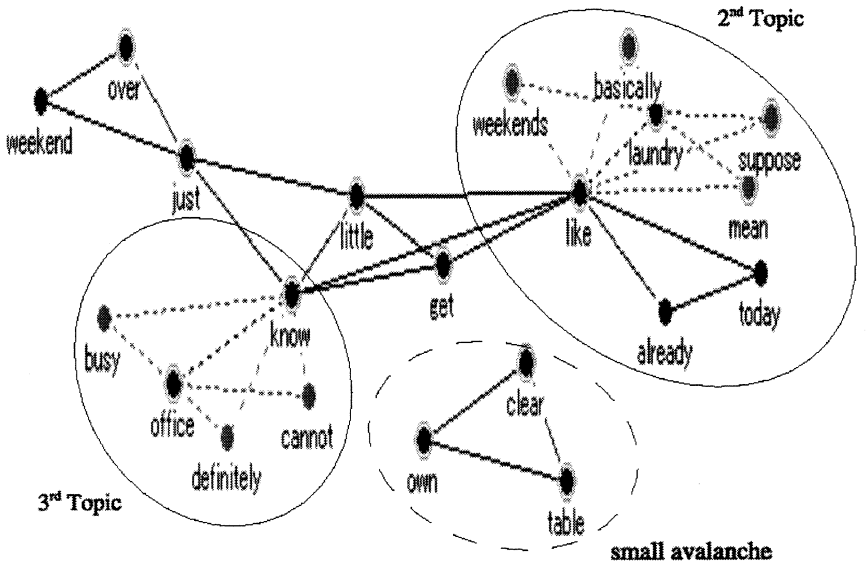


Fig. 4. Second & third crashes: Graphic view.

similar to the structure of the first crash: frequent words (black dots) surrounded by less frequent yet interrelated words (grey dots).

In order to explain this behavior we regard the theory of information, where less frequent or rare events carry higher amount of information. Back to the sandpile

paradigm, we could argue that the grains of sand that cause the pile to avalanche or to span, being rare events, carry higher amount of information. In the same tradition, the grey dots in Figs. 3 and 4, giving detailed information regarding the black dots they are surrounding, posses higher amount of information, as rare events, showing the tendency of criticality of human–human chat.

3.3. *Considerations*

In the experiment shown above, the *KeyGraph* tool was used rawly. In other words, preprocess of the chat was not performed. The utterances given by one of the speakers in one turn were considered as one sentence and were analyzed by words using the tool. In order to determine deeper relationship between the utterances of a critically self-organized chat flow a more detailed analysis of the dynamic behavior of the chat flow is needed, e.g. preprocessing the chat as for extracting high-frequency phrases, identifying relationship between low-frequency phrases and crucial utterances, etc. However, as fundamental idea, the above procedure showed to be satisfactory.

During the experiment the Zipf's law was used for a small database, which can be considered as the cause of the slope -3.64. Another influential factor is that the analyzed data is *spoken language*, therefore containing a lot of irregular words — back-channels and the like — making the results vary from the written English results — for a large database, the slope of the graph tends to be approximately 1.1. However, the tendency of critically self-organizing of the chat could be perceived. In the same regard, observing the dynamics of the chat, the positive feedback effects that these dynamics contain might be considered as triggering the self-organization. For example, interlocutors are interested in a given topic or opinion, making the chat grow toward a certain direction. However, a sudden unexpected event may change the growth trend. The change may begin slowly, but turns into a drastic revolution at a certain moment, giving birth to a new topic's trend, and then the cycle repeats itself until the chat section is finished. This behavior obeying the Zipf's law, showing the chat's tendency to critically self-organize.

4. Conclusion

In this paper, the discovery of CSOC is proposed. Understanding the critical behavior of a chat flow could improve modeling chat for computers in order to achieve a human-like computer chat. It was shown that discovering the moments when an utterance that may change the flow of the chat — a critical utterance — may appear could be crucial for keeping a smooth chat interaction. The Zipf's law was applied to the chat, been confirmed that the chat had self-organized to the critical point. An experiment was given, using the *KeyGraph* tool as a mean for visualizing the dynamic critical structure of the crashes. Future works are oriented toward the application of the Zipf's law to larger database in order to determine the frequency of not only words but also expressions used during human chat, and the

used of Chance Discovery techniques in order to determine the links between those expressions — and critical utterances — as for realizing the practical implementation of the discovery.

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